

Mitochondrial morphology in mitophagy and macroautophagy.

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Mitochondria are critical organelles in energy conversion, metabolism and amplification of signalling. They are however also major sources of reactive oxygen species and when dysfunctional they consume cytosolic ATP. Maintenance of a cohort of healthy mitochondria is therefore crucial for the overall cell fitness. Superfluous or damaged organelles are mainly degraded by mitophagy, a selective process of autophagy. In response to the triggers of mitophagy, mitochondria fragment: this morphological change accompanies the exposure of "eat-me" signals, resulting in the engulfment of the organelle by the autophagosomes. Conversely, during macroautophagy mitochondria fuse to be spared from degradation and to sustain ATP production in times of limited nutrient availability. Thus, mitochondrial shape defines different types of autophagy, highlighting the interplay between morphology of the organelle and complex cellular responses. This article is part of a Special Issue entitled: Mitochondrial dynamics and physiology.

Mitochondrial remodelling supports migration in white-crowned sparrows (*Zonotrichia leucophrys*).

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The migratory movements undertaken by birds are among the most energetically demanding behaviours observed in nature. Mitochondria are the source of aerobic energy production on which migration depends, but a key component of mitochondrial function, mitochondrial remodelling, has not been investigated in the context of bird migration. We measured markers of mitochondrial remodelling in the skeletal muscles of the Gambel's (migratory) and Nuttall's (non-migratory) white-crowned sparrows within and outside migratory periods. Gambel's were collected in (i) a non-migration period (baseline), (ii) preparation to depart for spring migration (pre-migration) and (iii) active autumn migration (mid-migration). Nuttall's were collected at timepoints corresponding to baseline and mid-migration in Gambel's. Across all sampling periods, we found that migratory birds had greater mitochondrial remodelling compared with non-migratory birds. Furthermore, birds from the migratory population also displayed flexibility, increasing several markers of mitochondrial remodelling (e.g. NRF1, OPA1 and Drp1) pre- and during migration. Further, the greater levels of mitochondrial remodelling and its upregulation during migration were specific to the pectoralis muscle used in flapping flight. Our study is the first to show that mitochondrial remodelling supports migration in Gambel's white-crowned sparrows, indicating a highly specific and efficient phenotype supporting the increased energetic demands of migration.

Pioneering Advances and Innovative Applications of Mesoporous Carriers for Mitochondria-Targeted Therapeutics.

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Mitochondria, known as the cell's powerhouse, play a critical role in energy production, cellular maintenance, and stemness regulation in non-cancerous cells. Despite their importance, using drug delivery systems to target the mitochondria presents significant challenges due to several barriers, including cellular uptake limitations, enzymatic degradation, and the mitochondrial membranes

themselves. Additionally, barriers in the organs to be targetted, along with extracellular barriers formed by physiological processes such as the reticuloendothelial system, contribute to the rapid elimination of nanoparticles designed for mitochondrial-based drug delivery. Overcoming these challenges has led to the development of various strategies, such as molecular targeting using cell-penetrating peptides, genomic editing, and nanoparticle-based systems, including porous carriers, liposomes, micelles, and Mito-Porters. Porous carriers stand out as particularly promising candidates as drug delivery systems for targeting the mitochondria due to their large pore size, surface area, and ease of functionalisation. Depending on the pore size, they can be classified as micro-, meso-, or macroporous and are either ordered or non-ordered based on both size and pore uniformity. Several methods are employed to target the mitochondria using porous carriers, such as surface modifications with polyethylene glycol (PEG), incorporation of targeting ligands like triphenylphosphonium, and capping the pores with gold nanoparticles or chitosan to enable controlled and triggered drug delivery. Photodynamic therapy is another approach, where drug-loaded porous carriers generate reactive oxygen species (ROS) to enhance mitochondrial targeting. Further advancements have been made in the form of functionalised porous silica and carbon nanoparticles, which have demonstrated potential for effective drug delivery to mitochondria. This review highlights the various approaches that utilise porous carriers, specifically focusing on silica-based systems, as efficient vehicles for targeting mitochondria, paving the way for improved drug delivery strategies in mitochondrial therapies.

RNA editing-induced structural and functional adaptations of NAD9 in *Triticum aestivum* under drought stress.

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Mitochondria are essential organelles in eukaryotic cells, producing ATP through the electron transport chain to supply energy for cellular activities. Beyond energy production, mitochondria play crucial roles in cellular signaling, stress responses, and the regulation of reactive oxygen species. In plants, mitochondria are one of the keys to responding to environmental stresses which can significantly affect crop productivity, particularly in crops like wheat. RNA editing, a post-transcriptional RNA modification process in mitochondria, is linked to regulating these stress responses. This study explores RNA editing patterns in the *nad9* gene of wheat drought-tolerant (Giza168) and drought-sensitive (Gemmiza10) wheat cultivars under drought stress to understand plant adaptation mechanisms. RNA-seq data for these cultivars were analyzed using CLC Genomic Workbench to identify RNA editing sites in the *nad9* gene, examining subsequent amino acid changes and predicting secondary structure modifications. These RNA editing sites were validated using qRT-PCR on drought-treated seedlings at 0, 2, and 12 hours post-treatment. Protein models were generated using AlphaFold, with functional predictions and structure verification conducted using various bioinformatics tools to investigate the effect of RNA editing on protein level. The results showed significant RNA editing events, especially C-to-T conversions, in the *nad9* gene across different drought exposure times. Giza168 had 22 editing sites, while Gemmiza10 had 19, with several showing significant differences between control and stress conditions. RNA editing influenced the NAD9 protein's secondary structure, particularly beta sheets, and 3D modeling highlighted the structural impacts of these edits. The N-terminal region of NAD9 contained important regulatory motifs, suggesting a complex regulatory environment. This study reveals key editing sites that differ between drought-tolerant and sensitive wheat cultivars, impacting NAD9 protein structures and highlighting the role of RNA editing in enhancing drought resilience. Additionally, the study suggests potential regulatory mechanisms, including phosphorylation and ubiquitination that influence mitochondrial stability and function.

Mitochondrial Ca(2+) uptake in skeletal muscle health and disease.

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Muscle uses Ca(2+) as a messenger to control contraction and relies on ATP to maintain the intracellular Ca(2+) homeostasis. Mitochondria are the major sub-cellular organelle of ATP production. With a negative inner membrane potential, mitochondria take up Ca(2+) from their surroundings, a process called mitochondrial Ca(2+) uptake. Under physiological conditions, Ca(2+) uptake into mitochondria promotes ATP production. Excessive uptake causes mitochondrial Ca(2+) overload, which activates downstream adverse responses leading to cell dysfunction. Moreover, mitochondrial Ca(2+) uptake could shape spatio-temporal patterns of intracellular Ca(2+) signaling. Malfunction of mitochondrial Ca(2+) uptake is implicated in muscle degeneration. Unlike non-excitable cells, mitochondria in muscle cells experience dramatic changes of intracellular Ca(2+) levels. Besides the sudden elevation of Ca(2+) level induced by action potentials, Ca(2+) transients in muscle cells can be as short as a few milliseconds during a single twitch or as long as minutes during tetanic contraction, which raises the question whether mitochondrial Ca(2+) uptake is fast and big enough to shape intracellular Ca(2+) signaling during excitation-contraction coupling and creates technical challenges for quantification of the dynamic changes of Ca(2+) inside mitochondria. This review focuses on characterization of mitochondrial Ca(2+) uptake in skeletal muscle and its role in muscle physiology and diseases.

Altered Ca2+ Handling and Oxidative Stress Underlie Mitochondrial Damage and Skeletal Muscle Dysfunction in Aging and Disease.

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Skeletal muscle contraction relies on both high-fidelity calcium (Ca2+) signals and robust capacity for adenosine triphosphate (ATP) generation. Ca2+ release units (CRUs) are highly organized junctions between the terminal cisternae of the sarcoplasmic reticulum (SR) and the transverse tubule (T-tubule). CRUs provide the structural framework for rapid elevations in myoplasmic Ca2+ during excitation-contraction (EC) coupling, the process whereby depolarization of the T-tubule membrane triggers SR Ca2+ release through ryanodine receptor-1 (RyR1) channels. Under conditions of local or global depletion of SR Ca2+ stores, store-operated Ca2+ entry (SOCE) provides an additional source of Ca2+ that originates from the extracellular space. In addition to Ca2+, skeletal muscle also requires ATP to both produce force and to replenish SR Ca2+ stores. Mitochondria are the principal intracellular organelles responsible for ATP production via aerobic respiration. This review provides a broad overview of the literature supporting a role for impaired Ca2+ handling, dysfunctional Ca2+-dependent production of reactive oxygen/nitrogen species (ROS/RNS), and structural/functional alterations in CRUs and mitochondria in the loss of muscle mass, reduction in muscle contractility, and increase in muscle damage in sarcopenia and a wide range of muscle disorders including muscular dystrophy, rhabdomyolysis, central core disease, and disuse atrophy. Understanding the impact of these processes on normal muscle function will provide important insights into potential therapeutic targets designed to prevent or reverse muscle dysfunction during aging and disease.

Virtual mitochondria: metabolic modelling and control.

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Inside the eukaryotic cell, mitochondria are internal organelles of prokaryotic origin, responsible for energy supply in the cell. The control of the mitochondrial ATP production is a complex problem with different patterns according to different tissues and organs. Our aim is to continue to develop the modelling of oxidative phosphorylation in different tissues, to model other parts of mitochondrial metabolism and to include this virtual mitochondria in a virtual cell. In constructing the complete metabolic map of mitochondria, we will take advantage of the sequenced genomes of eukaryotic organism (10-15% of the yeast genome concerns mitochondria).

Mitochondria in health, disease, and aging.

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Mitochondria are well known as organelles responsible for the maintenance of cellular bioenergetics through the production of ATP. Although oxidative phosphorylation may be their most important function, mitochondria are also integral for the synthesis of metabolic precursors, calcium regulation, the production of reactive oxygen species, immune signaling, and apoptosis. Considering the breadth of their responsibilities, mitochondria are fundamental for cellular metabolism and homeostasis. Appreciating this significance, translational medicine has begun to investigate how mitochondrial dysfunction can represent a harbinger of disease. In this review, we provide a detailed overview of mitochondrial metabolism, cellular bioenergetics, mitochondrial dynamics, autophagy, mitochondrial damage-associated molecular patterns, mitochondria-mediated cell death pathways, and how mitochondrial dysfunction at any of these levels is associated with disease pathogenesis. Mitochondria-dependent pathways may thereby represent an attractive therapeutic target for ameliorating human disease.

Central roles of apoptotic proteins in mitochondrial function.

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Mitochondria have been classically characterized as organelles with responsibility for cellular energy production in the form of ATP, but they are also the organelles through which apoptotic signaling occurs. Cell stress stimuli can result in outer membrane permeabilization, after which mitochondria release numerous proteins involved in apoptotic signaling, including cytochrome c, apoptosis-inducing factor, endonuclease G, Smac/DIABLO and Omi/HtrA2. Cell fate is determined by signaling through apoptotic proteins within the Bcl-2 (B-cell lymphoma 2) protein family, which converges on mitochondria. Many cancerous cells display abnormal levels of Bcl-2 protein family member expression that results in defective apoptotic signaling. Alterations in bioenergetic function also contribute to cancer as well as numerous other disorders. Recent evidence indicates that several pro-apoptotic proteins localized within mitochondria, as well as proteins within the Bcl-2 protein family, can influence mitochondrial bioenergetic function. This review focuses on the emerging roles of these proteins in the control of mitochondrial activity.

Human mitochondrial nucleases.

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Mitochondria are cytosolic organelles that have many essential roles including ATP production via oxidative phosphorylation, apoptosis, iron-sulfur cluster biogenesis, heme and steroid synthesis, calcium homeostasis, and regulation of cellular redox state. One of the unique features of these organelles is the presence of an extrachromosomal mitochondrial genome (mtDNA), together with all the machinery required to replicate and transcribe mtDNA. The accurate maintenance of mitochondrial gene expression is essential for correct organellar metabolism, and is in part dependent on the levels of mtDNA and mtRNA, which are regulated by balancing synthesis against degradation. It is clear that although a number of mitochondrial nucleases have been identified, not all those responsible for the degradation of DNA or RNA have been characterized. Recent investigations, however, have revealed the contribution that mutations in the genes coding for these enzymes has made to causing pathogenic mitochondrial diseases.
